

GALVESTON BUOY 42035, TX, USA

WMO: 996970

Lat: 29.232N

Lon: 94.413W

Elev: 0

StdP: 101.33

Time Zone: -6.00 (W06)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.2	(c) 5.2	(d) -4.0	(e) 2.7	(f) 6.5	(g) -1.8	(h) 3.3	(i) 8.0	(j) 14.7	(k) 8.1	(l) 14.0	(m) 8.4	(n) 8.5	(o) 350	(p) 0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 1.9	(c) 30.6	(d) 26.0	(e) 30.2	(f) 26.0	(g) 30.0	(h) 25.9	(i) 27.4	(j) 29.4	(k) 27.0	(l) 29.2	(m) 26.8	(n) 29.1	(o) 3.7	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.4	28.9	26.4	21.8	28.8	26.0	21.4	28.7	86.3	29.4	84.9	29.3	83.7	29.2	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.5	10.4	DB	1.5	31.4	2.2	0.8	-0.1	32.0	-1.3	32.4	-2.6	32.8	-4.1	33.4
			WB	-0.6	27.7	2.1	1.1	-2.1	28.5	-3.3	29.2	-4.5	29.8	-6.0	30.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	12.6	13.4	16.9	20.6	24.3	27.6	28.4	28.8	27.2	23.9	18.8	14.7
	DBStd	6.35	3.51	3.58	3.09	2.39	1.88	1.10	0.95	1.01	1.40	2.66	3.47	3.65
	HDD10.0	41	18	13	2	0	0	0	0	0	0	0	1	7
	HDD18.3	543	177	138	61	8	1	0	0	0	0	2	36	121
	CDD10.0	4229	99	108	216	317	444	528	570	584	516	430	266	151
	CDD18.3	1691	0	1	16	76	187	278	312	326	266	173	50	7
	CDH23.3	16491	0	0	1	66	1104	3158	3881	4197	2909	1138	36	0
(13)	CDH26.7	4941	0	0	0	1	22	864	1439	1749	782	84	0	0
(14)	Wind	WSAvg	5.8	6.0	6.0	5.6	6.2	5.3	5.1	4.7	5.6	6.0	6.2	6.1
(15) Precipitation	PrecAvg	1437	123	87	96	99	124	140	138	124	150	133	105	119
	PrecMax	2001	276	193	213	229	238	412	358	269	358	397	268	259
	PrecMin	757	16	26	19	10	1	16	32	19	36	12	17	32
	PrecStd	301	66	53	50	66	70	86	79	80	72	114	60	53
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	20.1	22.9	25.8	28.2	30.7	30.6	31.6	30.2	28.8	25.0	22.3
		MCWB	18.6	19.6	21.6	23.5	23.7	26.3	25.8	26.5	25.5	25.3	23.2	21.4
	2%	DB	18.2	19.1	21.9	24.7	27.2	29.9	30.2	30.8	29.7	28.2	24.3	21.4
		MCWB	17.8	18.6	21.2	23.0	24.5	25.6	25.9	26.1	25.6	24.9	22.0	20.6
	5%	DB	17.7	18.4	21.2	24.1	26.8	29.4	29.9	30.5	29.5	27.7	23.7	20.7
		MCWB	17.2	18.0	20.4	22.9	24.3	25.4	25.9	26.1	25.4	24.7	21.7	19.7
	10%	DB	17.0	17.7	20.5	23.5	26.5	29.0	29.6	30.2	29.1	27.0	23.0	19.8
		MCWB	16.3	17.0	19.7	22.1	24.3	25.2	25.9	26.0	25.2	24.1	21.3	18.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	20.1	22.4	24.6	26.4	27.9	27.5	28.0	27.7	26.8	24.1	21.8
		MCDB	18.9	20.1	22.5	24.9	26.5	29.4	29.5	30.0	28.7	27.4	25.0	22.0
	2%	WB	18.1	18.9	21.6	23.9	25.9	27.1	27.0	27.5	26.9	26.1	23.0	20.9
		MCDB	18.1	19.0	21.8	24.3	26.3	28.7	29.3	29.8	28.8	26.9	23.6	21.3
	5%	WB	17.5	18.2	20.8	23.4	25.4	26.6	26.8	27.1	26.5	25.6	22.4	20.2
		MCDB	17.6	18.3	21.1	23.8	26.1	28.5	29.2	29.7	28.8	26.8	23.2	20.7
	10%	WB	16.7	17.4	20.2	22.7	25.0	26.2	26.6	26.8	26.1	25.0	21.8	19.1
		MCDB	16.9	17.6	20.5	23.2	25.9	28.3	29.1	29.6	28.6	26.6	22.8	19.7
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.5	2.8	2.3	1.8	1.7	1.8	1.9	2.2	2.7	3.4	3.8
		MCDBR	3.5	3.1	2.2	2.4	1.7	1.8	1.7	2.2	2.1	2.4	3.2	3.3
	5% WB	MCWBR	4.2	3.4	2.5	2.9	2.0	1.9	1.6	1.9	1.9	2.6	4.1	4.1
		MCDWR	3.3	2.9	2.0	2.3	1.5	1.7	1.7	1.9	2.2	2.4	3.2	3.4
(40) Clear-Sky Solar Irradiance	taub		0.352	0.365	0.380	0.431	0.475	0.461	0.483	0.466	0.446	0.389	0.375	0.357
		taud	2.521	2.494	2.441	2.299	2.211	2.315	2.274	2.324	2.337	2.484	2.466	2.512
	Ebn at Noon		886	903	908	868	827	833	814	826	832	865	849	860
		Edh at Noon	90	100	111	133	145	130	135	128	122	99	94	87
(44) All-Sky Solar Radiation	RadAvg		2.93	3.35	4.64	5.56	6.33	6.51	6.11	5.83	5.03	4.49	3.41	2.62
	RadStd		0.29	0.54	0.40	0.42	0.39	0.45	0.53	0.45	0.53	0.53	0.34	0.29

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+97

Nomenclature: See separate page